ecological features that support distinct or important ecological communities at a regional scale are present. The Queensland and Marion Plateaux, together support over 20 coral reefs and cays, which provide complex habitats with diverse and abundant invertebrate and fish communities. Similarly, the northernmost parts of the Tasmantid Seamount Chain contain a diverse range of habitats, including deep-water sponge gardens and near-pristine tropical coral

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SOURCE: MEPC.268(68).pdf
communities. Similarly, the northernmost parts of the Tasmantid Seamount Chain contain a diverse range of habitats, including deep-water sponge gardens and near-pristine tropical coral reef systems. Collectively these are known to be biological hot-spots, with significant species diversity.

1.3 Over millennia, the geological and oceanographic history of the region and its warm and cool current patterns have prevented the migration of species, prompting the development of flora and fauna that evolved, adapted and spread in isolation. Localized currents can act as a barrier to dispersal; a s a result the area has high levels of species endemism.

1.4 The area contains some of the world's most unique and globally significant marine species, such as the leatherback turtle (Dermochelys coriacea) , humphead Maori wrasse (Cheilinus undulatus) and nautilus (Nautilus pompilius).

1.5 The Coral Sea provides migratory corridors for cetaceans, sharks, fish, turtles and seabirds, many of which are of conservation concern. Successive research efforts have highlighted the significance of the Coral Sea in patterns of dispersal, whereby the reefs

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seabirds, many of which are of conservation concern. Successive research efforts have highlighted the significance of the Coral Sea in patterns of dispersal, whereby the reefs provide a series of dispersal stepping-stones from the western Pacific towards the GBR.

1.6 Deep, cold water troughs and abyssal basins are habitat for an array of benthic species, many of which are a protected matter under Australias Environment Protection and Biodiversity Conservation Act 1999 (EPBC Act) or international agreement or are unique to the area, including 18 endemic species of deep-water sharks, rays and chimaera fish.

1.7 The remoteness and challenging environment of the area means much of its biodiversity remains undescribed and new species found nowhere else are routinely discovered. Surveys in the deeper reef habitats at Osprey Reef have revealed prehistoric six-gilled sharks, giant oil fish and many crustaceans and unidentified fish at depths of 1400 metres. A unique, dwarf speciation of Nautilus pompilius was also identified on Osprey Reef. This species evolved isolated from nautilus in the nearby Coral Sea and GBR, and is a

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1400 metres. A unique, dwarf speciation of Nautilus pompilius was also identified on Osprey Reef. This species evolved isolated from nautilus in the nearby Coral Sea and GBR, and is a reflection of the endemic nature of ecosystems within the area.

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
1.8 The north-western Coral Sea hosts the only confirmed spawning aggregation of black marlin (Makaira indica) in the world. This species migrates throughout the Pacific Ocean, but only uses the Coral Sea to spawn.

1.9 The few detailed surveys undertaken have shown that as many as 40 5% of the invertebrates inhabiting seamounts in the area are new to science while up to 34% of the species may be endemic. Scientists expect that research into the lesser known, deeper areas of the region will uncover many new species.

Critical habitat

1.10 The area contains outstanding examples of reef communities and a diverse array of

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species may be endemic. Scientists expect that research into the lesser known, deeper areas of the region will uncover many new species.

Critical habitat

1.10 The area contains outstanding examples of reef communities and a diverse array of isolated sandy cays, islands, deep-sea plains, seamounts and canyons. Collectively, these areas provide foraging, breeding and nesting grounds for a rich array of marine species, including 341 species that are recognized for their conservation significance under the EPBC Act and under international agreements. This includes 26 species of cetaceans, 219 species of corals, 21 species of fish, 46 species of sharks and rays, five species of marine turtles and 24 species of birds. Many of these species are listed as threatened or migratory species, or both, and, whilst over half of these species show declining population trends worldwide, many are still found at healthy levels in the Coral Sea.

Dependency

1.11 The reefs of Coral Sea provide stepping-stones for the dispersal of species between the GBR and the greater Pacific Ocean region. Maintaining the overall integrity and resilience

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Dependency

1.11 The reefs of Coral Sea provide stepping-stones for the dispersal of species between the GBR and the greater Pacific Ocean region. Maintaining the overall integrity and resilience of these reefs is therefore necessary to ensure that they can function effectively as stepping-stone habitats.

1.12 The Coral Sea also provides migratory corridors for cetaceans, sharks, fish, turtles and seabirds, many of which are of conservation concern. There are likely to be further important areas for feeding, breeding, migrating and resting that have yet to be clearly identified, and which may act as critical habitat for many species.

1.13 Thousands of species rely on the ecosystem processes within the Coral Sea region to provide opportunities for foraging, breeding and migration. These processes are largely driven by the availability of energy within the system, which in turn is dependent upon the unique interactions between the region's oceanographic and topographic features.

1.14 Localized turbulence in the lee of reefs, islands and seamounts influences biological communities by creating patches of high to intermediate productivity in the nutrient-poor

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SOURCE: MEPC.268(68).pdf
1.14 Localized turbulence in the lee of reefs, islands and seamounts influences biological communities by creating patches of high to intermediate productivity in the nutrient-poor open-water environment. These sites of enhanced productivity are important aggregators for a range of pelagic species including small fish, mid-trophic predatory fish and large predators. They are also known to attract a range of species of conservation concern including marine turtles, marine mammals and seabirds.

1.15 Cays in the area offer important habitat for seabirds to roost and nest, and turtles to lay eggs above the high tide mark. Seabirds provide the main source of energy on these cays, through their nutrient-rich guano, eggs and carrion, which support food webs of terrestrial and intertidal invertebrates and over time facilitate the development of soil and organic matter, which in turn provides habitat for more complex plant communities.

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(adopted on 15 May 2015)
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https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
1.16 The species assemblage and trophic structure of the region relies on the highly interlinked web of local oceanographic patterns, topography and energy inputs. An impact on any one of these can disrupt this web, destabilizing ecosystem processes and the species that rely on them.

Representativeness

1.17 The reefs, cays and herbivorous fish of the Queensland and Marion Plateaux and

the northern extent of the Tasmantid Seamount Chain highlight the biological significance and ecological value of the region.

1.18 The Queensland and Marion Plateaux are considered important for aggregations of marine life and the high levels of biodiversity they support. The reefs and islands of these plateaux are approximately 200 400 kilometres from the coast and provide for diverse and abundant invertebrate (e.g. sea cucumber) and fish communities.

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plateaux are approximately 200 400 kilometres from the coast and provide for diverse and abundant invertebrate (e.g. sea cucumber) and fish communities.

1.19 These reefs of the Queensland and Marion Plateaux are known for their particularly high densities of shark species, and Osprey Reef in particular is also recognized for its populations of the iconic humphead Maori wrasse and nautilus. The lagoons of these reefs are important nursery sites for sharks and predatory fish, while the island areas support critical nesting sites for the green turtle and a range of seabird species. The plateaux also abut two significant deep-water regions: the Queensland Trough, which separates the Queensland Plateau from the GBR, and the Townsville Trough, which separates the two plateaux from each other. These troughs contain canyons and gullies that are likely to support unique deep-water ecosystems.